

STUDENT PERFORMANCE ANALYTICS

Project Report and Documentation

1. Introduction

Student Performance Analytics is a Python-based project designed to analyze student academic information and classify students according to their overall performance. The implementation loads a student dataset, cleans the data, calculates an overall score, creates performance categories, produces visualizations, and applies a Decision Tree classification model.

2. Objectives

- To load and examine student performance data.
- To clean missing and duplicate records.
- To calculate an overall performance percentage.
- To categorize students as Excellent, Good, Average, or Needs Improvement.
- To analyze attendance, study hours, and academic marks.
- To visualize student performance using graphs.
- To train a machine-learning model for performance prediction.
- To predict the performance category of a new student.

3. Dataset

The program uses a CSV file named **student_performance.csv**. The source code uses fields including Student_ID, Age, Gender, Study_Hours, Attendance, Assignment_Score, Internal_Marks, Previous_Marks, and Final_Marks. The dataset is loaded using Pandas.

4. Methodology

1. Load the CSV dataset.
2. Inspect columns, missing values, and statistical information.
3. Remove duplicate records and handle missing values.
4. Calculate Overall_Score using assignment, internal, and final marks.
5. Create performance categories based on the calculated score.
6. Perform descriptive analysis and identify top-performing students.
7. Generate performance visualizations.
8. Select machine-learning features and target.
9. Split the data into training and testing sets.
10. Train a Decision Tree Classifier.
11. Evaluate the model using accuracy and a classification report.
12. Use the model to predict a new student's performance.
13. Save the processed dataset.

5. Performance Calculation

Overall Score = Assignment Score × 0.20 + Internal Marks × 0.30 + Final Marks × 0.50

Category	Condition
Excellent	Score ≥ 80
Good	$60 \leq \text{Score} < 80$
Average	$40 \leq \text{Score} < 60$
Needs Improvement	Score < 40

6. Algorithm

Decision Tree Classification

A Decision Tree Classifier is used to predict the student's performance category. The implementation uses Age, Study_Hours, Attendance, Assignment_Score, Internal_Marks, and Previous_Marks as input features, while Performance is the target variable. The model is configured with max_depth=5 and random_state=42.

7. Data Splitting and Model Evaluation

The dataset is divided using an 80:20 training/testing split. The source code uses stratified splitting and random_state=42. After training, the model predicts the test records. Model performance is evaluated using accuracy_score and classification_report.

8. Implementation

The project is implemented in Python using Pandas and NumPy for data handling, Matplotlib for visualization, and Scikit-learn for machine learning. The program also creates a processed_student_performance.csv file after completing the analysis.

9. Visualizations

- Attendance vs Final Marks scatter plot.
- Student Performance Distribution bar graph.
- Average Final Marks by Performance Category bar graph.

The provided source code specifies these visualizations, but no actual screenshot images or numeric output results were included in the supplied source-code document. Therefore, this report does not invent graph values or model accuracy.

10. Results

The system produces student performance categories, identifies the top 10 students, displays descriptive statistics, generates graphs, evaluates the Decision Tree model, predicts the performance of a sample new student, and saves the processed dataset. The exact model accuracy is produced when the program is executed on the actual student_performance.csv dataset.

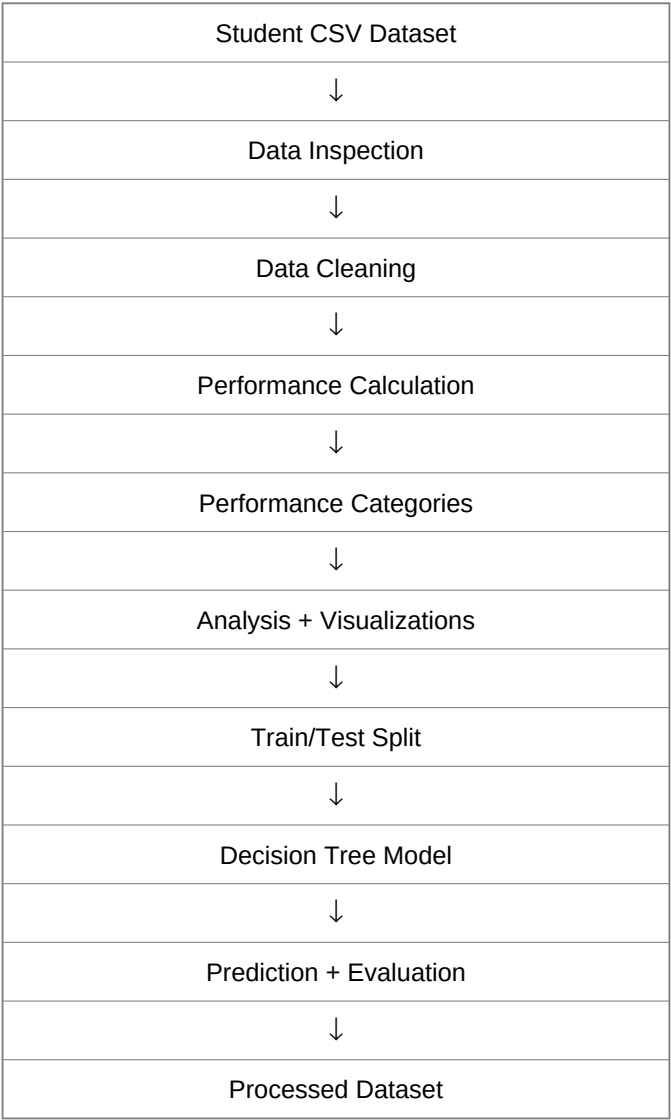
11. Sample New Student Prediction

Input	Value
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Age	20
Study Hours	3.5
Attendance	85
Assignment Score	78
Internal Marks	72
Previous Marks	75

The trained model uses these values to predict the student's performance category.

12. Project Flow Diagram



13. Conclusion

The Student Performance Analytics project provides a complete workflow for analyzing student academic data. It combines data cleaning, performance calculation, visualization, and machine learning. The Decision Tree model allows the system to classify and predict student performance based on selected academic and attendance-related features. The project can be extended with a dashboard, database connectivity, additional machine-learning models, and automated reports.

14. Source Basis

This report is prepared from the supplied Student Performance Analytics source code. The source specifies the dataset file, preprocessing steps, performance formula, categories, visualizations, Decision Tree model, sample prediction, and processed-data output.